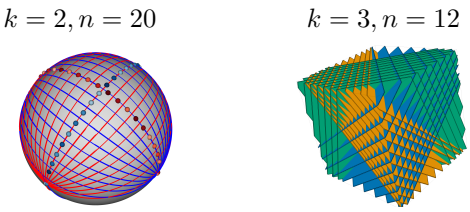

Figure 6. **Example geometries of linearly compositional models. Left:** 2 concepts ($n = 20$ each) on a sphere. Each colored stripe is the argmax boundary for one concept value; their intersections yield 20^2 combination cells. **Right:** 3 concepts ($n = 12$ each) in 3D. Colored planes show argmax boundaries; their intersections carve out 12^3 combination cells. Each boundary is colored according to the concept it belongs to.

orthogonal structure is a plausible target that current vision(-language) systems *should* exhibit if they are to generalize compositionally.

Takeaway §4.2. In the binary GD+CE regime, linear factorization with cross-concept orthogonality is both necessary and sufficient for compositional generalization.

4.3. Packing and minimum dimension

So far we have established necessary and sufficient conditions for the representation space $\mathcal{Z}$ to exhibit compositional generalization. However, it is not clear what exactly the capacity constraints are on the representation space to support it. Here, we ask a basic capacity question: what is the minimum embedding dimension d needed to support Divisibility (Desideratum 1), i.e. realize all possible n^k combinations? The following result gives a tight lower bound. Proof and its sketch in Section E.

Proposition 3 (Minimum dimension for linear probes)**.** For k concepts, each with n values, suppose there exist linear probes that correctly classify each concept value for all n^k combinations from embeddings $f(\boldsymbol{x}) \in \mathbb{R}^d$. Then necessarily $d \geq k$.

Importantly, the bound is independent of the number of values n per concept, depending only on the number of concepts k. This holds regardless of whether each factor is discrete or continuous, since the proof requires only that any two values per factor be distinguishable. We illustrate two examples of divisibility in Fig. 6: on a sphere and in Euclidean space, though our formal results establish minimal dimensionality only for Euclidean space. Additional visualizations in Fig. 21. As the number of concepts k increases while the embedding dimension d is fixed, divisibility imposes increasingly tight packing constraints: factors within each concept must lie in progressively lower-dimensional subspaces, approaching near-collinearity in the limit (an example in Fig. 16 in Appendix).

Empirical results and the dependence on representation space and the loss function. The result in Proposition 3 is a geometric lower bound and does not depend on a specific loss or representation space. Empirically, we find that CE in Euclidean space reaches this bound most closely, BCE typically requires higher dimension (approximately $2k$), and spherical geometry adds roughly one extra dimension (see the setting in Section I.6, and Fig. 40 for the empirical results). In Section F.2 we discuss an idealized case where a model's logits are constrained to take two values, yet satisfy perfect classification, and show that this necessarily requires $d \geq 1 + k(n-1)$ in Proposition 6.

Takeaway §4.3. The minimum embedding dimension scales with the number of concepts k, not the number of values n per concept ($d \geq k$). When k grows relative to a fixed d, per-concept subspaces must become increasingly low-rank, approaching near-collinearity.

5. Surveying necessary conditions in pretrained models

Previous section established necessary conditions for compositional generalization: representation space must be linearly factorized and the factors must be orthogonal across concepts (Proposition 1). In this section, we ask: how far are modern pretrained models from these necessary conditions?

Our theory is built on binary concept values, but some of the concepts in the datasets we consider are multi-valued. Instead of repeatedly sampling binary values and testing factorization on these, we adopt the natural multivalued extension of the necessary structure. Concretely, we test whether representations admit an approximate additive decomposition of the form

$$\boldsymbol{z_c} \approx \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i}, \quad (\boldsymbol{u}_{i,a} - \boldsymbol{u}_{i,b}) \quad \perp (\boldsymbol{u}_{j,a'} - \boldsymbol{u}_{j,b'}), \quad (6)$$

for all $\boldsymbol{c} \in [n]^k$, all $i \neq j$, and all $a, b, a', b' \in [n]$, i.e., an additive per-concept factorization with cross-concept orthogonality of difference directions. This form reduces to the binary case when $n = 2$.

Main qualitative result. We give intuition for both conditions in (6) using a pretrained DINOv3 model on dSprites (Fig. 7). The figure shows that the idealized linearity and orthogonality conditions are visible in the global PCA view **(a)**, and that the same pattern is observed when some concepts are fixed and others are varied **(b)**, consistent with approximate additive factorization and cross-concept orthogonality. Any datapoint can approximately be expressed as a sum of the factor vectors $\boldsymbol{u}$ **(c)**, and they often take low-rank structure, as well as satisfy orthogonality **(d)**. The remainder of this section quantifies these observations. For full qualitative results, see Section I.5.

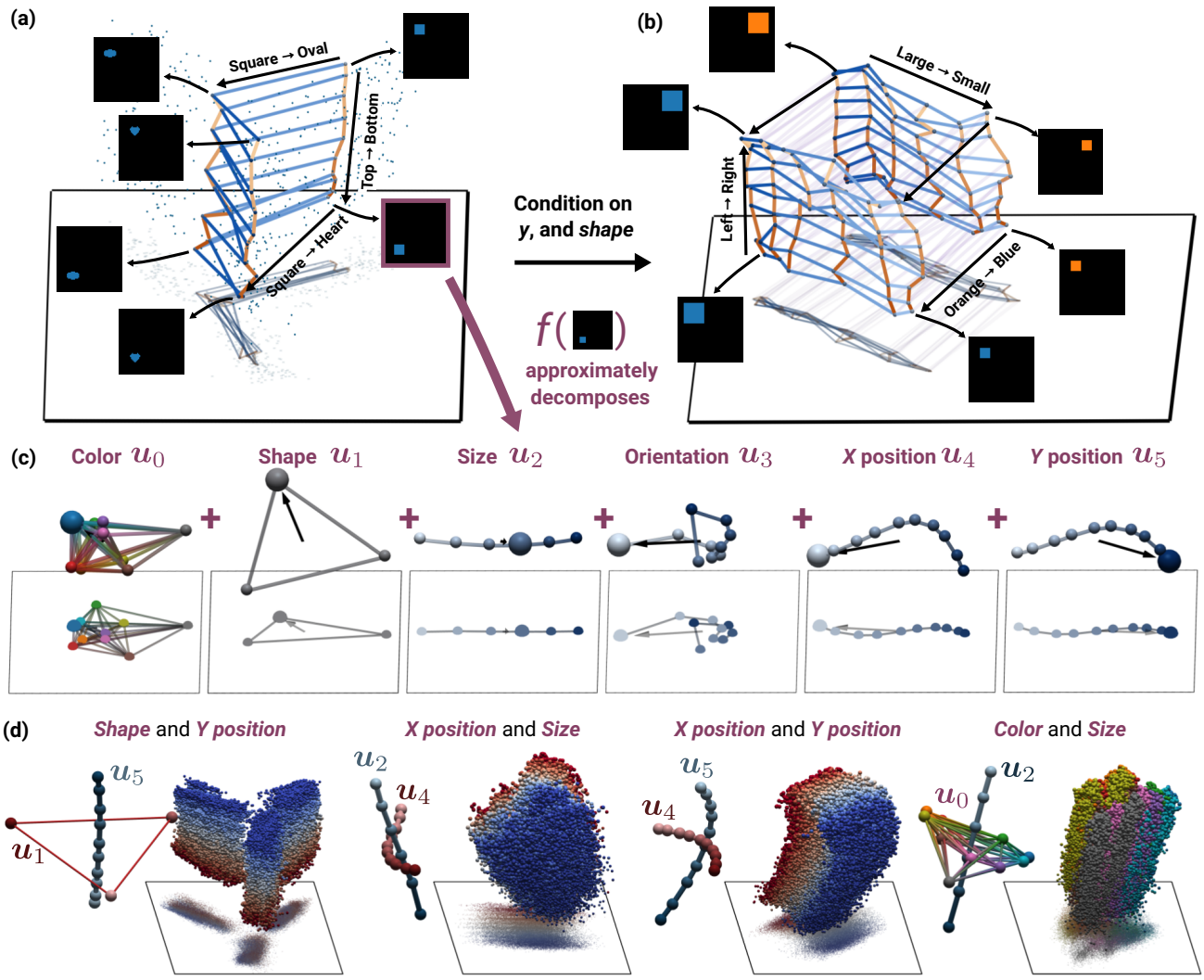

Figure 7. **Summary of the linearity + orthogonality hypothesis, illustrated in DINOv3 on dSprites. (a)** PCA projection of DINOv3 embeddings over all dSprites combinations. Changing shape or y-position produces near-constant direction shifts (linearity), and the two directions are nearly orthogonal. **(b)** After fixing shape to square and y to one value, PCA on the remaining subset shows that horizontal position x (left $\rightarrow$ right), size (large $\rightarrow$ small), and color (orange $\leftrightarrow$ blue) each vary along near-constant directions and form a grid-like structure. **(c)** Embeddings exhibit approximate linear factorization: each embedding decomposes as a sum of one factor per concept, $\boldsymbol{z_c} \approx \sum_i \boldsymbol{u}_{i,c_i}$ (illustrated for the highlighted sample, with arrows pointing to the selected factor in each panel). The recovered factors are typically low-rank ($\leq$ 3D), so these 3D plots capture most of their structure. **(d)** For each pair of concepts, the left mini-panel shows the two sets of factors, illustrating near-orthogonality across concepts. The right mini-panel shows all datapoints projected onto the span of those two factors; the projected points organize into a grid aligned with the factor directions, consistent with additive decomposition. The corresponding quantitative results are reported in Sections 5.1, 5.3 and 5.4; full qualitative results in Section I.5.

Concretely, we quantitatively evaluate the necessary conditions for compositional generalization in pretrained models. We aim to answer the following questions:

Q4 (Section 5.1) *Is linear factorization present in pretrained models?*

Q5 (Section 5.2) *Does the degree of linear factorization correlate with compositional generalization?*

Q6 (Section 5.3) *Are per-concept difference vectors approximately orthogonal across concepts, as the theory predicts?*

Q7 (Section 5.4) *What geometric structure do factors $\boldsymbol{u}$ exhibit?*

Models and datasets. We evaluate a broad model set spanning contrastive vision-language and self-supervised vision encoders: OpenAI CLIP (Radford et al., 2021), OpenCLIP, MetaCLIP (Xu et al., 2023), MetaCLIP2 (Chuang et al., 2025) checkpoints from the LAION ecosystem (Schuhmann et al., 2021a), SigLIP and SigLIP 2 (Zhai et al., 2023; Tschannen et al., 2025), and DINOv1–v3 (Caron et al., 2021; Siméoni et al., 2025). This covers different architectures,

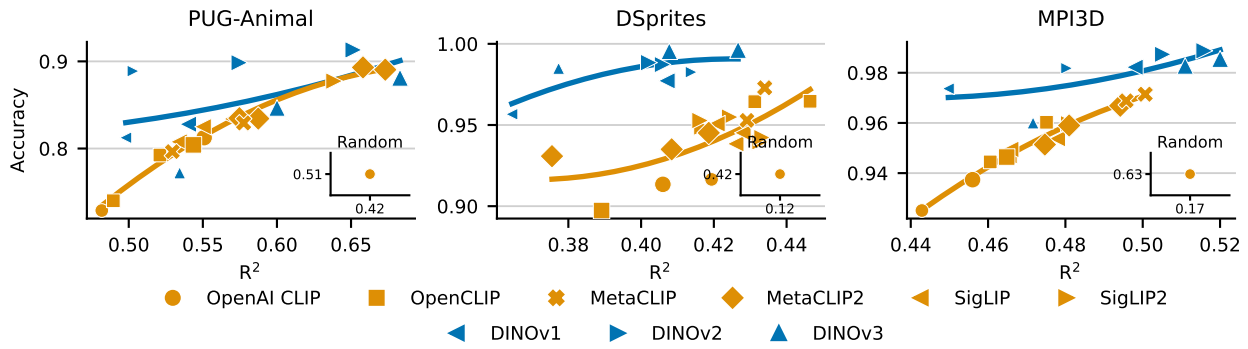

Figure 8. **Linearity in embeddings correlates with compositional generalization across VLMs and self-supervised vision models.** Across three datasets, we plot compositional generalization accuracy against projected R^2 (our linear-factorization score) for a broad set of pretrained encoders, including vision-language models (OpenAI CLIP, OpenCLIP, SigLIP, MetaCLIP, MetaCLIP2) and pure vision backbones (DINOv1–v3). Each marker is a model variant; higher projected R^2 is consistently associated with higher compositional generalization performance.

training objectives (softmax vs. sigmoid), and scales. We evaluate on three compositional datasets: `PUG-Animal` (Bordes et al., 2023), `dSprites` (Matthey et al., 2017), and `MPI3D` (Gondal et al., 2019), plus ImageNet-AO (Abbasi et al., 2024) (Section I.4.2). The full checkpoint roster, dataset summary, and example samples are listed in Section H.5.

Recovering the factors from representations. Assuming that a linear factorization exists in the representations of a model f as detailed in Section 4.1, we can recover the factors $\{\boldsymbol{u}_{i,j}\}_{i\in[k],j\in[n]}$ by averaging over all the datapoints that share a particular concept value (Trager et al., 2023). For analysis purposes it is sufficient to recover the centered factors. That is, given all centered embeddings $\{f(\boldsymbol{x}_{\boldsymbol{c}})\}_{\boldsymbol{c}\in[n]^k}$, the factors can be recovered as $\boldsymbol{u}_{i,j} = \frac{1}{|\{\boldsymbol{c}\in[n]^k : c_i = j\}|}\sum_{\boldsymbol{c}\in[n]^k : c_i=j} f(\boldsymbol{x}_{\boldsymbol{c}})$.

5.1. Linear factorization in pre-trained models

To assess the extent of linearity in the embeddings, we measure a whitened R^2 score on the probe span. We (1) project onto the probe span to discard information the embeddings may possess beyond the dataset concepts, and (2) whiten the embedding space so that R^2 is not inflated by a few dominant directions. Concretely, given the recovered approximate factors $\{\boldsymbol{u}_{i,j}\}_{i\in[k],j\in[n]}$, the R^2 score is computed as

$$R^2 = 1 - \frac{\sum_{\boldsymbol{x}_{\boldsymbol{c}}\in\mathcal{D}} \left\| f(\boldsymbol{x}_{\boldsymbol{c}}) - \sum_{i=1}^{k} \boldsymbol{u}_{i,c_i} \right\|_2^2}{\sum_{\boldsymbol{x}_{\boldsymbol{c}}\in\mathcal{D}} \| f(\boldsymbol{x}_{\boldsymbol{c}}) - \bar{\boldsymbol{f}} \|_2^2}, \tag{7}$$

where $\mathcal{D}$ is the dataset, and $\bar{\boldsymbol{f}}$ is the mean embedding. Note that a score of 1.0 indicates perfect linearity. We provide intuition of linear factorization and its relation to the R^2 in Section H.3, additional justification of whitening in Section H.2.

Results. Fig. 9 shows projected R^2 scores across models and datasets. Among all datasets, each model's R^2 score is consistently above the random baseline (0.4-0.6 vs. 0.12-0.42, respectively). This suggests that embeddings are partially captured by a sum of per-concept components, while still leaving some information unexplained. Additionally, we observe that R^2 scores are similar in scale across models. The same linearity trend holds in the zero-shot setting when using text encoders as probes on both PUG-Animal and ImageNet-AO (Section I.4 and Figs. 28 and 32).

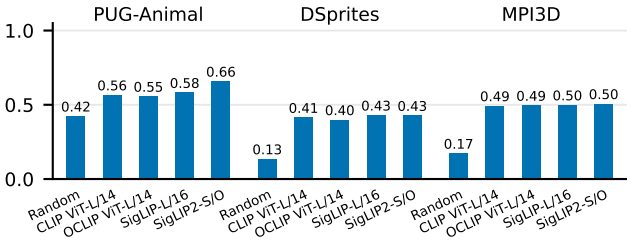

Figure 9. **Linear factorization partly explains current models' embedding spaces.** Bar plots of whitened R^2 on three datasets with varying concept/value counts.

Importantly, we note that the R^2 scores, while consistently above random, are far from perfect, indicating that current models only partially satisfy the linear factorization predicted by our theory.

Takeaway §5.1. Embeddings exhibit partial linear factorization (R^2 typically 0.4–0.6), explaining a moderate fraction of the variance via per-concept components. The gap from perfect scores highlights a divergence from the ideal embedding structure theory predicts.

5.2. Compositional generalization and linear factorization

We ask whether the *degree* of linear factorization predicts compositional generalization.

Metrics and setup. For each dataset/model, we train linear probes on 10% of all concept combinations and evaluate